

Read p. 306 of textbook.

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, we say this is an indeterminate form.

Different indeterminate forms:

$$\pm \frac{\infty}{\infty}, \frac{0}{0}, \infty - \infty, 0 \times \infty$$

$$1^\infty, 0^0, \infty^0 \text{ — Indeterminate powers.}$$

Not indeterminate forms: $\frac{\infty \times \infty}{\infty}, \frac{\infty + \infty}{\infty}$

When you have an indeterminate form, use L'Hospital's rule:

If $\lim_{x \rightarrow * } \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ or $\frac{\pm \infty}{\pm \infty}$,

where $*$ = $\infty, -\infty$ or $a \in \mathbb{R}$, then

$$\lim_{x \rightarrow *} \frac{f(x)}{g(x)} = \lim_{x \rightarrow *} \frac{f'(x)}{g'(x)}$$

Examples:

$$1) \lim_{x \rightarrow \infty} \frac{x}{1+x^2} \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{1}{2x} = 0$$

$$2) \lim_{x \rightarrow \infty} \frac{3^x}{x^2} \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{3^x \ln 3}{2x} \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{3^x \ln 3 \times \ln 3}{2} = \infty$$

$$3) \lim_{x \rightarrow -\infty} \frac{2^x}{x^2} = 0$$

$$\left(2^{-\infty} = \frac{1}{2^{\infty}} \right)$$

$$4) \lim_{x \rightarrow \infty} \frac{\ln x}{x^2} \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{2x} = \lim_{x \rightarrow \infty} \frac{1}{2x^2} = 0.$$

$$5) \lim_{x \rightarrow -\infty} \frac{\ln x}{x^2} \text{ Not defined.}$$

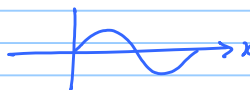
$$6) \lim_{x \rightarrow 0} \frac{2x}{e^{3x} - 1} \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{2}{3e^{3x}} = \frac{2}{3}.$$

$$7) \lim_{x \rightarrow \pi/2} \frac{\cos x}{2x - \pi} \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow \pi/2} \frac{-\sin x}{2}$$

$$= \frac{-\sin \pi/2}{2} = -\frac{1}{2}.$$



$$8) \lim_{x \rightarrow \infty} \frac{x \ln x}{x + \ln x} \quad \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{L'Hop}{=} \lim_{x \rightarrow \infty} \frac{1 \cdot \ln x + 1}{1 + \frac{1}{x}} \quad \left(\frac{\infty}{1+0} \right)$$

$$= \infty$$

$$9) \lim_{x \rightarrow \infty} \frac{x^2}{4 + e^x}$$

Form $\infty - \infty$

• Factor

• Add fractions.

Examples:

$$1) \lim_{x \rightarrow \infty} x^4 - x^3 \quad (\infty - \infty)$$

$$= \lim_{x \rightarrow \infty} x^3(x - 1) \quad (\infty \times \infty)$$

$$= \infty$$

$$2) \lim_{x \rightarrow -\infty} x^4 + x^3 \quad (\infty - \infty)$$

$$= \lim_{x \rightarrow -\infty} x^3(x + 1) \quad (-\infty \times -\infty)$$

$$= +\infty$$

$$3) \lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$$

(Textbook p. 309) NB

$$= \lim_{x \rightarrow 1^+} \frac{x-1 - \ln x}{(x-1) \ln x}$$

(add fractions)

Examples: Form $\infty \times 0$

• Write as a fraction

$$1) \lim_{x \rightarrow \infty} x e^{-x} \quad (\infty \times 0)$$

$$= \lim_{x \rightarrow \infty} \frac{x}{e^x} \quad \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

$$2) \lim_{x \rightarrow \infty} x e^{\frac{3}{x}} \quad (\infty \times e^0) \quad (\infty \times 1)$$

$$= \infty$$

$$3) \lim_{x \rightarrow \infty} \frac{1}{x} \ln x \quad (0 \times \infty)$$

$$= \lim_{x \rightarrow \infty} \frac{\ln x}{x} \quad \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

Indeterminate pairs (p. 810)

Examples:

Form 0^0 ∞^0 1^∞

$$1) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^{3x} \quad (\text{This limit is of the form } 1^\infty)$$

$$\therefore \text{Start as follows: Let } y = \left(1 + \frac{1}{x} \right)^{3x}$$

$$\therefore \ln y = \ln \left(1 + \frac{1}{x} \right)^{3x} = 3x \ln \left(1 + \frac{1}{x} \right)$$

Now determine $\lim_{x \rightarrow \infty} \ln y$

$$2) \lim_{x \rightarrow \infty} (e^x + 6x)^{\frac{2}{x}} \quad (\infty^0)$$

$$\text{Let } y = (e^x + 6x)^{\frac{2}{x}} \quad (\text{we want } \lim_{x \rightarrow \infty} y)$$

$$\ln y = \ln (e^x + 6x)^{\frac{2}{x}} = \frac{2}{x} \ln (e^x + 6x)$$

$$= \frac{2 \ln (e^x + 6x)}{x}$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{2 \ln (e^x + 6x)}{x} \quad \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{\frac{2}{e^x + 6x} \times (e^x + 6)}{1}$$

$$= \lim_{x \rightarrow \infty} \frac{2(e^x + 6)}{e^x + 6x} \quad \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{2(e^x)}{e^x + 6} \quad \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{2e^x}{e^x} = 2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \ln y = 2 \quad (\text{we want } \lim_{x \rightarrow \infty} y)$$

$$\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln y} = e^{\lim_{x \rightarrow \infty} \ln y}$$

$$= e^2$$

$$3) \lim_{x \rightarrow 0^+} (2x)^{\frac{1}{\ln x}} \quad (0^0)$$

$$\text{Let } y = (2x)^{\frac{1}{\ln x}} \quad (\text{I want: } \lim_{x \rightarrow 0^+} y)$$

$$\begin{aligned} \ln y &= \ln (2x)^{\frac{1}{\ln x}} \\ &= \frac{1}{\ln x} \ln 2x \\ &= \frac{\ln 2x}{\ln x} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 0^+} \ln y &= \lim_{x \rightarrow 0^+} \frac{\ln 2x}{\ln x} \quad \left(\frac{-\infty}{-\infty} \right) \\ &\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow 0^+} \frac{\frac{1}{2x} \times 2}{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0^+} 1 = 1 \end{aligned}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \ln y = 1 \quad (\text{I want } \lim_{x \rightarrow 0^+} y)$$

$$\Rightarrow \lim_{x \rightarrow 0^+} y = e^1 = e.$$